Practical No.3

Aim: Message Authentication Codes:
Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

Code for implementing MD5 Algorithm :

```
import hashlib
result = hashlib.md5 (b'Priya')
result1 = hashlib.md5 (b'Rajak')
# printing the equivalent byte value.
print ("The byte equivalent of hash(Priya):", end ="")
print(result.digest ())
print ("The byte equivalent of hash(Rajak) :", end ="")
print (result1.digest () )
```

Output:

```
========== RESTART: E:\INS\MD5.py ====
The byte equivalent of hash(Priya):b'q\xe9\x1c\xbc}\x97u\x8b_a\x92nNh,\x12'
The byte equivalent of hash(Rajak) :b'f\x06\x07\xb7[\n\xe1\x1c\xe6%\xfd\xb4I\xe1\xdep'
```

Code for implementing SHA Algorithm :

```
import hashlib
text = input("Enter the value to encode: ")
result = hashlib.sha1(text.encode())
print("The Hexadecimal Equivalent of SHA1 is:")
print(result.hexdigest())
```

Output:

```
============================ RESTART: E:\INS\SHA.py
Enter the value to encode priyarajak
The Hexadecimal Equivalent if SHA1 is:
aba01ce8a9d5bca1f416cbf0e8a6f4d5b8b3ac2c
```

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